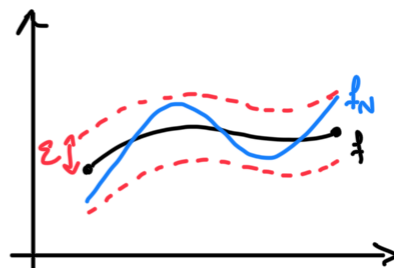


Last time: $f_n: X \rightarrow Y$ converges uniformly to $f: X \rightarrow Y$

if: $\forall \varepsilon > 0 \exists N \in \mathbb{N}$ s.t.

$$d_Y(f_n(x), f(x)) < \varepsilon \quad \forall n \geq N$$

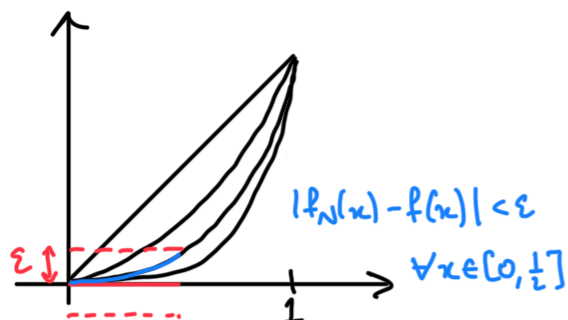
and $\forall x \in X$.



Ex $f_n(x) = x^n$, $f(x) = \begin{cases} 0 & x \in [0, 1) \\ 1 & x = 1 \end{cases}$

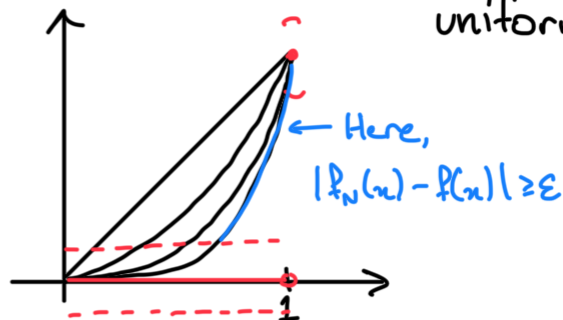
① $f_n, f: [0, \frac{1}{2}] \rightarrow \mathbb{R}$:

f_n converges uniformly to f



② $f_n, f: [0, 1] \rightarrow \mathbb{R}$:

f_n does not converge uniformly



Thm Let X, Y be metric spaces and let $f_n, f: X \rightarrow Y$ be functions. If f_n is continuous $\forall n \in \mathbb{N}$ and $\{f_n\}_{n \geq 1}$ converges uniformly to f , then f is continuous.

Pf: Fix $x_0 \in X$ and $\varepsilon > 0$. Want: $\exists \delta > 0$ s.t.

$$d_X(x, x_0) < \delta \Rightarrow d_Y(f(x), f(x_0)) < \varepsilon.$$

As f_n converges uniformly to f , $\exists N$ s.t.

$$d_Y(f_n(x), f(x)) < \frac{\varepsilon}{3} \quad \forall n \geq N \text{ and } \forall x \in X.$$

As f_N is continuous at x_0 , $\exists \delta > 0$ s.t.

$$d_X(x, x_0) < \delta \Rightarrow d_Y(f_N(x), f_N(x_0)) < \frac{\varepsilon}{3}.$$

Altogether,

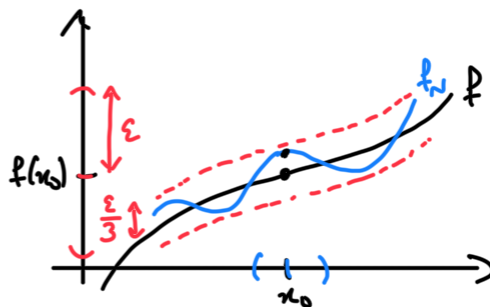
$$d_X(x, x_0) < \delta$$

$$\Rightarrow d_Y(f(x), f(x_0))$$

$$\leq d_Y(f(x), f_N(x))$$

$$+ d_Y(f_N(x), f_N(x_0)) + d_Y(f_N(x_0), f(x_0))$$

$$< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$$



as desired. □

Def Let X be a metric space. We define

$$C(X) = \{f: X \rightarrow \mathbb{R} : f \text{ is continuous and bounded}\}$$

$$d_{\text{sup}}(f, g) = \sup \{|f(x) - g(x)| : x \in X\}$$

Rmk ① The sup exists since f, g are bounded.

② If X is also compact, then by the extreme value theorem,

$$C(X) = \{f: X \rightarrow \mathbb{R} : f \text{ is continuous}\}.$$

Exc d_{sup} is a metric on $C(X)$.

Prop Let X be a metric space and $f_n, f \in C(X)$.

Then: $\{f_n\}_{n \geq 1}$ converges uniformly to $f \iff$

$f_n \rightarrow f$ in the metric space $(C(X), d_{\text{sup}})$, i.e.

$$d_{\text{sup}}(f_n, f) = \sup_{x \in X} |f_n(x) - f(x)| \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Pf: $\Rightarrow$: Suppose $\{f_n\}$ converges uniformly to f .

Want: $d_{\sup}(f_n, f) \rightarrow 0$ as $n \rightarrow \infty$. Fix $\varepsilon > 0$.

By uniform convergence, $\exists N$ s.t.

$$|f_n(x) - f(x)| < \frac{\varepsilon}{2} \quad \forall n \geq N \text{ and } \forall x \in X$$

$$\Rightarrow \sup_{x \in X} |f_n(x) - f(x)| \leq \frac{\varepsilon}{2} \quad \forall n \geq N$$

$$\stackrel{''}{=} d_{\sup}(f_n, f)$$

$$\Rightarrow d_{\sup}(f_n, f) \leq \frac{\varepsilon}{2} < \varepsilon \quad \forall n \geq N.$$

So $d_{\sup}(f_n, f) \rightarrow 0$ as $n \rightarrow \infty$.

$\Leftarrow$: Suppose $d_{\sup}(f_n, f) \rightarrow 0$ as $n \rightarrow \infty$. Want:

$\{f_n\}$ converges uniformly to f . Fix $\varepsilon > 0$. As

$d_{\sup}(f_n, f) \rightarrow 0$, $\exists N$ s.t.

$$d_{\sup}(f_n, f) < \varepsilon \quad \forall n \geq N$$

$$\stackrel{''}{=} |f_n(x) - f(x)| \leq \sup_{x \in X} |f_n(x) - f(x)|$$

$$\Rightarrow |f_n(x) - f(x)| < \varepsilon \quad \forall n \geq N \text{ and } \forall x \in X$$

So $\{f_n\}$ converges uniformly to f . $\square$

Ex $f_n(x) = x^n$

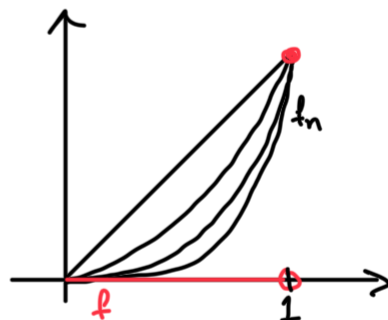
① $f_n \in C([0, 1])$: since $f_n: [0, 1] \rightarrow \mathbb{R}$ is continuous $\forall n \geq 1$.

② $\{f_n\}_{n \geq 1}$ does not converge in $C([0, 1])$:

Suppose not: $f_n \rightarrow f$ in $C([0, 1])$. Then f_n

converges uniformly to f . In particular, f_n converges pointwise to f , and so (by Lecture 33)

$$f(x) = \begin{cases} 0 & x \in [0, 1) \\ 1 & x = 1 \end{cases}$$



But

$$d_{\text{sup}}(f_n, f) = \sup_{x \in [0, 1]} |x^n - f(x)| \geq 1$$

for any $n \geq 1$. This contradicts that $f_n \rightarrow f$ in $C([0, 1])$.

Thm Let X be a metric space. Then the metric space $(C(X), d_{\text{sup}})$ is complete.

Pf: Fix a Cauchy sequence $\{f_n\}$ in $C(X)$, i.e.:
 $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t.

$$d_{\text{sup}}(f_n, f_m) < \varepsilon \quad \forall m, n \geq N.$$

For each $x \in X$, note that $\{f_n(x)\}$ is a Cauchy sequence in $\mathbb{R}$ since

$$|f_n(x) - f_m(x)| \leq \sup_{x \in X} |f_n(x) - f_m(x)| = d_{\text{sup}}(f_n, f_m).$$

As $\mathbb{R}$ is complete, $\exists$ some $f(x) \in \mathbb{R}$ s.t. $f_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$.

Claim: $\{f_n\}$ converges uniformly to f . Fix $\varepsilon > 0$.

As $\{f_n\}$ is Cauchy, $\exists N$ s.t.

$$\sup_{x \in X} |f_n(x) - f_m(x)| = d_{\text{sup}}(f_n, f_m) < \frac{\varepsilon}{2} \quad \forall n, m \geq N.$$

So, for each $x \in X$,

$$|f_n(x) - \underbrace{f_m(x)}_{\rightarrow f(x) \text{ as } m \rightarrow \infty}| < \frac{\varepsilon}{2} \quad \forall m, n \geq N$$

$$\Rightarrow |f_n(x) - f(x)| \leq \frac{\varepsilon}{2} \quad \forall n \geq N$$

$$\left(\text{By HW4: } \begin{array}{l} a_n \rightarrow a \text{ in } \mathbb{R} \\ a_n \leq M \quad \forall n \geq N \end{array} \right\} \Rightarrow a \leq M$$

In other words,

$$|f_n(x) - f(x)| < \varepsilon \quad \forall n \geq N \text{ and } \forall x \in X.$$

So $\{f_n\}$ converges uniformly to f .

(To be continued !)